

EDUCATION

University of Michigan, Ann Arbor, MI

[Graduation: December 2026]

Bachelor of Science - Mathematics || *Bachelor of Science* - Statistics || *Minor* - Economics

- U-M Quantitative and Computational Finance Group (QCF) - Quant Developer - (Statistical Arbitrage team)
- U-M MIG Annual Quant Competition - **Rank: 1st** place champion/ ± 120 participants
- U-M Society of Undergraduate Math Students - Events Coordination Staff and Member

Programming Languages: || Python, R, C++ , Matlab || **Libraries:** SciKit-Learn, Pandas, QuantLib, NumPy || **Learning:** SQL ||

Relevant Coursework:

❖ **Mathematics:**

Real Analysis, Linear Algebra, Probability Theory, Differential Geometry, Abstract Algebra
Differential Equations and Dynamical Systems, Financial Mathematics, Mathematical Modelling, Stochastic Processes, Numerical Analysis*, Stochastic Analysis*, Measure Theory*, Combinatorics*

❖ **Statistics + Data Science:**

Regression Analysis, Machine Learning, Statistical Computing, Theoretical Statistics, Bayesian Inference*

❖ **Computer Science:**

Programming and Data Structures, Discrete Mathematics

❖ **Economics:**

Microeconomic Theory, Macroeconomic Theory, Game Theory, Econometrics*

(*indicates courses audited/self taught, not taken for credit as of May 2025)

RESEARCH EXPERIENCE

Math Researcher, University of Michigan Computational Medicine, Ann Arbor, MI

[May 2025 - Present]

Topic: Optimal Transport Theory and Cellular Reprogramming

- Investigated optimal transport frameworks to model cell identity transformations as probabilistic mass shifts between gene expression distributions
- Applied Wasserstein metrics and entropic regularization to characterize the most efficient transformation between cellular states

Math Researcher, University of Michigan Math Department, Ann Arbor, MI

[August 2024 - Present]

Topic: Large Scale Geometries of the Integers

- Developed algorithms to compute word-lengths of the Word Metric on the 4×4 Heisenberg Group and over the Integers $\mathbb{Z}$.
- Currently developing algorithms to prove the isomorphism between the group of Rough Isometries $RI(G, d)$ and $\mathbb{Z}/2\mathbb{Z}$.

Math Researcher, University of Michigan Math Department, Ann Arbor, MI

[September 2023 - November 2023]

Topic: Smooth k -Manifolds and their Applications

- Studied connections between behaviors of locally-Euclidean manifolds, and their applications to relativistic mechanics, applied calculus, and data analysis.

Physics Researcher, Washtenaw Community College Physics Department, Ann Arbor, MI

[August 2021- December 2021]

Topic: Noether's Invariance Theorems and their Significance to Modern-day Conservation Laws

- Developed a simple, rigorous construction of Noether's theorem's applications into conservation laws with numerical methods.

WORK EXPERIENCE

Garuda Metalindo, Jakarta, Indonesia

[June 2024- July 2024]

Data Analyst and Internal Audit Intern

- Engineered C++ code and Python scripts to streamline data processing for inventory management of over 2.5 billion products and 50,000 variants annually.

Michigan Math Laboratory

[September 2023 - Present]

Tutor and Gateway Testing Proctor

- Facilitated an all-inclusive and personalized teaching approach for over 700 students a single week by teaching calculus and matrix algebra in tailored one-on-one sessions.

Deloitte Indonesia, Jakarta, Indonesia

[February 2018- March 2018]

Risk Advisory High School Intern

- Reviewed risk database reports for compliance and fraud prevention in current and past bank clients.
- Applied statistical methods to analyze historical client data, identifying patterns of risk exposure across market, credit, and operational domains.
- Assisted development of HR risk assessment frameworks aimed at reducing exposure to employee-related liabilities and compensation claims.